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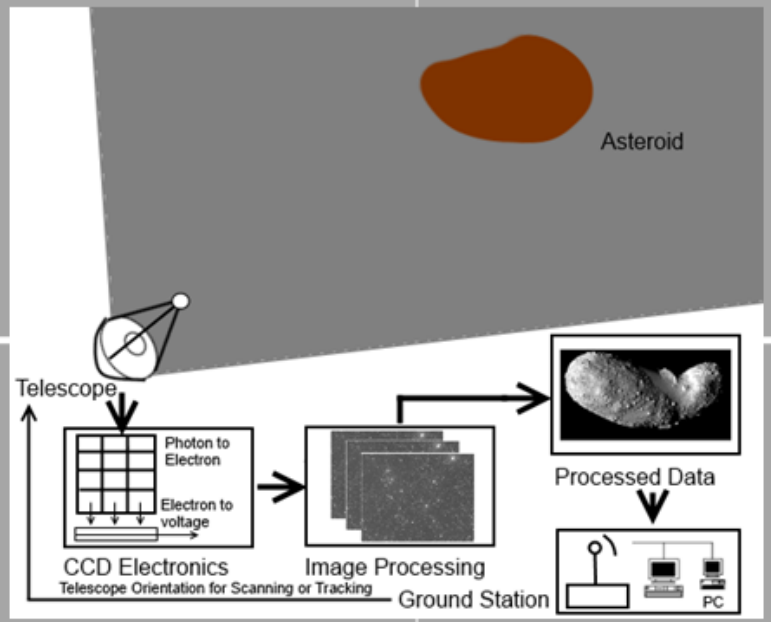
2. Overview Chart

Title and Research Team

- Dr. David Hyland, PI, Texas A&M University
- Hyerim Kim, Graduate Student, Texas A&M University
- Roy Tucker, CoI for Electronics, ECAPS LLC
- Shen Ge, CoI for System Engineering, ECAPS LLC
- Dr. Neha Satak, CoI for DSP, ECAPS LLC

Research Objectives

- Detect and characterize NEAs smaller than 1000 m diameter which are large enough to cause impact damage but small enough that most have not been discovered.
  - Develop autonomous imaging processing algorithm for use in small spacecrafts.
  - Design scab mode imaging (TDI) based space telescope for detecting near-earth asteroids.
  - Refine orbital parameters of known asteroids by tracking them.
  - Start at TRL = 1 and end at TRL = 3 since simulations verify design.



Approach

- Design a space-based telescope built upon heritage ground-based technology.
- Develop on-board imaging algorithms to reduce communication requirements.
- Test algorithms via comprehensive simulations for NEA detection probability.

Potential Impact

- Enable design of a cost-effective space-based telescope to detect potentially hazardous asteroids in regions that are difficult or impossible to observe from Earth.
- Permit acquisition of follow-up observations of NEAs previously discovered from ground-based facilities

3. Scientific/Technical/Management

a. Relevance of the proposed research to the specific STRO-ESI solicitation goals, objectives and topics

The work proposed herein relates to the NASA exploration goals by helping develop *"technologies to enable detection and characterization of NEOs, with an emphasis on near-Earth asteroids, using small satellites or CubeSats, along with the on-board computational algorithms that minimize the data transmission requirements back to Earth (STRO-ESI 2013, section I(d), Topic 1)."* Specifically, the Space Moving Object and Transient Event Search System (SMOTESS) answers two of the three explicit goals of Topic 1 of NASA STRO-ESI. SMOTESS provides the clear answer to an "*innovative, small satellite-based measurement technology [ies] that would enable a new generation of observational capabilities for detecting and remotely characterizing NEOs"* and *"efficient on-board identification and characterization computational algorithms that minimize the data required to be transmitted back to Earth for processing (STRO-ESI 2013, section I(d), Topic 1)."*

b) Technical Approach and Methodology

NASA is seeking technologies to enable detection and characterization of near-Earth objects (NEOs) using small satellites along with on-board computational algorithms that minimize the data transmission requirements back to Earth. Texas A&M University and Experimental Center for Applied Physical Systems LLC (henceforth referred to as ECAPS) together will achieve three research objectives of effective asteroid detection, tracking, and data processing through the development of the Space Moving Object and Transient Event Search System (SMOTESS).

<u>Research Objective 1: Detection</u>

The SMOTESS instrument is a configuration of four small telescopes (one for redundancy). It does not require accurate attitude tracking as in typical space telescope designs since it uses the scan mode imaging technique which has been demonstrated on-ground in the existing heritage system MOTESS [7]. The MOTESS has detected numerous NEAs, with objects ranging from asteroids with diameters about 300 meters such as 1997 MW1 down to asteroids with diameters of tens of meters such as 2003 UY12.

The basic SMOTESS instrument is a copy of the MOTESS instrument and consists of three 14" f/5 paraboloidal mirrors with a 2048x2048 CCD with 12 micron square pixels at the prime focus. The focal length will be 175 cm which produces an image scale of 1.414 arcseconds per pixel. At an average distance of 0.7 AU (Venus-like orbit) with a 24-hour scan rate, SMOTESS has a limiting magnitude detection threshold which is at least 23 with regard to solar elongation as shown in Table 1.

Table 1: Various limiting magnitudes of SMOTESS with respect to solar elongation at a Venus-like orbit [6].

Solar Elongation (deg)	30	40	60	90	140	180
Detection Threshold (magnitude)	23.0	23.4	23.9	24.2	24.4	24.4

Assuming a lifetime of ten years and a field of view in the rotation plane of 300° (everywhere except within 30° of the sun) and 15° above and below the rotation plane, SMOTESS is estimated to be able to see ~120,000 asteroids of size between 40 to 140 meter diameters if placed in a Venus-like orbit. Figure 1 shows the percentage and number of asteroids detectable with respect to the asteroid size distribution estimated by our preliminary models based upon magnitudes and distributions generated by Bottke et al. and Greenstreet et al. respectively [3, 5].

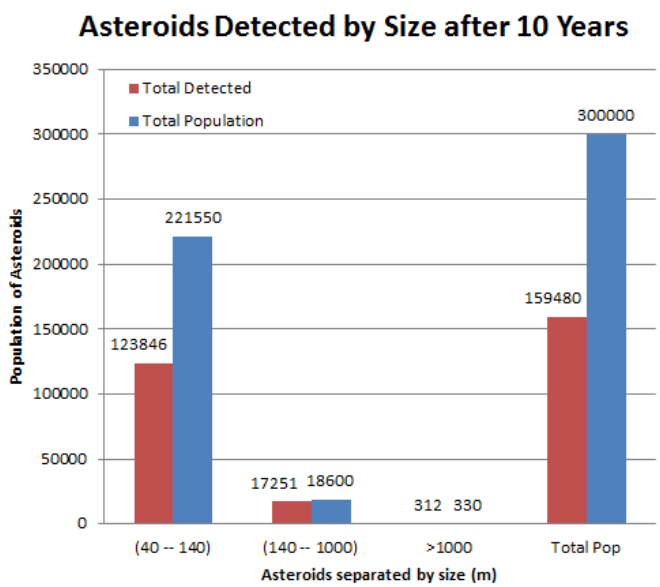


Figure 1: Percentage and number of asteroids detectable with respect to the asteroid size distribution.

The population of total detectable asteroids as well as the 40 - 140 meter diameter family of asteroids with regard to the lifetime of SMOTESS is shown in Figure 2. As can be seen, a steady increase in detection progresses throughout the lifetime of SMOTESS.

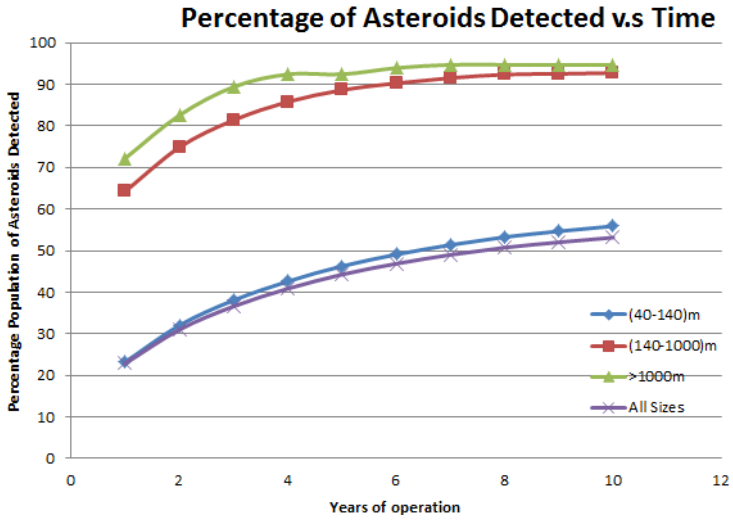


Figure 2: Number of asteroids in each respective time detectable with regard to time in years.

Research Objective 2: Data Processing

The SMOTESS instrument will be capable of processing the sky images on-board and sending the astrometric positions of the identified asteroids to the ground. It will be capable of identifying new asteroids autonomously and efficiently using parallel computing (GPUs) with an upgraded version of the demonstrated asteroid identification algorithm at MIT Lincoln Lab which has been used for over twenty years.

The research objective is to modify the ground-based MIT Lincoln lab's algorithm to reduce the computational power required onboard the spacecraft while still achieving reliable asteroid detections. This will be achieved by using the information available from the spacecraft's basic attitude determination in the image processing algorithm for asteroid detection. The algorithm will be enhanced to reduce false detections during observation in space based environment. Complimentary to the software development is designing reliable hardware with low power requirements for implementing the asteroid detection algorithm onboard the spacecraft.

Research Objective 3: Tracking

Using the Minor Planet Center (MPC)'s database of asteroids, the optimal latitude to direct the SMOTESS instrument can be determined to continually orient the spacecraft to track newly discovered asteroids by moving its telescope configuration to cover the predicted paths of the asteroids. The telescope configuration will be optimized for using the minimum energy to reorient which will maximize the lifetime of the science mission.

Follow-up observations from ground-based observatories are hard since many NEAs can only be discovered when they are close to the Earth. Trying to refine the orbital parameters of such objects are very slow processes unless they can be pinged with radar. The ability to observe these

objects in very different parts of their orbits permits great refinements of their orbits. The SMOTESS instrument on a spacecraft in a Venus-like orbit enables this capacity.

The SMOTESS payload on a spacecraft at a Venus-like orbit will allow it to see much of the near-earth asteroid population in the solar plane currently undetectable or detectable with great difficulty from Earth. Figure 3 is the System Block Diagram showing how the design works together. First, the telescope captures light from an asteroid. This is sent to the scan-mode CCD array which produces 12 GB of images over a span of 12 hours. These images are processed through on-board image processing software that searches the images for asteroids, detects them, and calculate their orbital trajectories. This processed data is then sent via antenna to a ground station on Earth. SMOTESS uses a TDI or scan mode imaging which offers a number of advantages over the more traditional stare mode imaging which most ground telescopes and space telescopes use. Features of scan mode imaging and stare mode imaging are compared in Table 2.

Table 2: Features of scan mode imaging and stare mode imaging.

Features of Scan Mode Imaging	**Features of Stare Mode Imaging**
Continuous imaging process allows no wasted photons and high instrument productivity but with limits on integration time	Total freedom to image any part of the sky with whatever integration time is desired
Simultaneous integration, read out, and movement to next field	Collection of photons must be interrupted when moving to the next field
Precision pointing and tracking not required but limited to a band around the sky near the rotational plane	Requires precision pointing and tracking
Reduction of CCD 'cosmetics' due to averaging effect of imaging process	
Simplified and improved image calibration	

Currently, the TRL of SMOTESS is at 1 and is expected to reach TRL 3 at the end of the research.

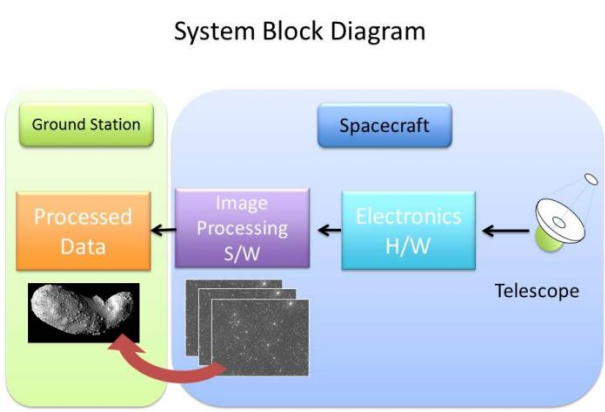


Figure 3: System block diagram showing the operational process of SMOTESS allowing near-earth objects to be detected.

Task 1: Mission Requirements

A basic model of the population of near earth asteroids (NEAs) need to be coded and defined so as to allow a prediction of the performance of the telescope for various configurations, i.e. the numbers of asteroids of various sizes detectable by SMOTESS at a Venus-like orbit. This is required to confirm the performance advantages of the SMOTESS. Though preliminary code has been generated, the code needs to be refined.

Task 2: Optical Design

The original MOTESS prototype consists of three 35 cm aperture f/5 Newtonian telescopes, each with a 1024 x 1024 CCD imager at the focus. The resulting image scale is 2.83 arcseconds per pixel for a field width of 48.3 arcminutes. Sitting stationary on the earth's surface, the system relies upon the earth's rotation to produce the scanning motion for the Time Delay Integration imaging technique that the cameras employ. The TDI or scan-mode imaging method forces rigid interrelated constraints on the optics, imager, instrument pointing, field-of-view, and equivalent integration time. Using the earth's rotation, the instrument described above has an equivalent integration time at the equator of 193.2 seconds. In spite of bright atmospheric skyglow which contributes about 20,000 photoelectrons per pixel, the system regularly reveals stellar sources on dark nights down to about magnitude 20.5.

The telescope supports are fabricated from welded pipe and plumbing fittings since there is no need for an accurately pointing and tracking mounting. Almost all of the money for the construction went into the optics and CCD cameras, those portions of the instrument that actually produce the images. Accurately pointing and tracking mountings are very expensive but collect no photons.

The starting point SMOTESS concept design consists of four 35 cm aperture f/5 telescopes, essentially a space-borne descendant of the original prototype. We describe this as a 'starting point' design because the scan-mode imaging technique imposes numerous interdependent compromises on the design and performance parameters. A single change in specified performance or hardware characteristic will necessitate a cascade of changes to ripple through everything else. Instead of a single design, we have a family of designs in multi-dimensional space with parameters of aperture, focal length, CCD array size, pixel size, equivalent integration time, scan rate, detection threshold, field of view, etc. As an example, if, along the way, we decide that we must trade some detection threshold for launch weight, we can change a couple of numbers in a spreadsheet and quickly determine the new instrument design parameters.

The unambiguous detection of moving objects requires a minimum of three images. We propose an array of four telescopes for a couple of reasons. First of all, if we were to lose one of three telescopes, the mission is concluded. Game over. A fourth telescope will not only provide redundancy, but it will also permit a lower detection threshold by processing an additional image.

We propose to use a very simple optical design for the telescopes, a concave paraboloidal mirror with the CCD imaging device at the focus. Such a simple design will minimize loss of light from multiple reflections and absorption in refractive elements, ghost images from reflections, reduce

weight, and reduce the mechanical complexity of aligning multiple elements and keeping them aligned during the rigors of launch.

One requirement of the scan-mode imaging technique is that the optical system must introduce minimal field distortions such as ‘pin-cushion’ or ‘barrel’ distortion. As stars drift across the surface of the CCD, they must do so in straight lines, strictly along the columns of the parallel registers. An optical system consisting only of a paraboloidal mirror will satisfy this condition at the cost of some off-axis coma. This off-axis aberration is to some extent helpful in that it will prevent single-pixel star images. Acquiring precise astrometric positions of stars and asteroids is facilitated by centroiding multipixel stars. The position of a single pixel star is only known to an accuracy of +/- 0.5 pixels whereas a multipixel star can be centroided to about +/- 0.2 to 0.1 pixels.

Task 3: Electronic Design

The CCD camera controller for each telescope is a relatively standard design providing clocking waveforms and correlated double-sampling (CDS) video processing. The additional requirement for scan-mode imaging is that the CCD clocking must be precisely coordinated with the rotation rate of the instrument platform. The clocking rate for a 24-hour rotation rate is a modest 24 kilopixels per second. A twelve-hour rotation rate will require a clocking rate twice that or about 48 kilopixels per second, not terribly challenging for low-noise CCD video processing. It is likely that the maximum rotation rate would be about once in six hours which leads to a clocking rate approaching 100 kilopixels per second. At that rate, the read noise begins to increase because of the shorter sampling times and this will begin to significantly affect the detection threshold.

Other than selecting electronic components rated for use in a space environment and designing-in redundancies to permit bypassing component failures, the design will be very low-risk

Task 4: Optical and Electronic Integration

As mentioned above, the clocking of the CCDs must be carefully coordinated with the rotation of the instrument scan platform in order to produce punctiform star images in the resulting imagery data. This rotation rate must be precisely measured in order to compute continuously the desired CCD clocking rate. It will also be necessary to determine the rotational axis of the scan platform and be able to make adjustments to its orientation. Significant deviations from the desired axis direction will cause the star images to drift across the instrument CCDs at some angle to the parallel register columns and produce elongated star images. In addition to control of the scan platform rotation axis, it will be highly desirable to implement the ability to adjust the rotational position of the CCD in the telescope to compensate for any residual star drift direction misalignment.

Additional telescope internal functions will include focus and collimation of the primary mirror. This will be achieved by means of three linear actuators separated by 120 degrees behind the mirror. They may be moved together to adjust focus or differentially to adjust collimation.

Finally, it will be desirable to control the pointing of the telescope with respect to the plane of rotation of the scan platform by an amount of about plus or minus ten to fifteen degrees.

Task 5: Image Processing & Detection

As mentioned previously, the onboard algorithm will be designed such that it can autonomously detect asteroids and transmit only the orbital elements of the detected asteroids back to earth. The image processing & detection algorithm has three main steps. The first step orients the CCD image on the celestial sphere using the known stars and their relative geometry found in the star catalog. The second step identifies potential asteroids from the images. The third step computes the astrometric positions of the identified asteroids and removes the false detections.

The orientation of an image from the CCD is found with respect to the celestial sphere by comparing the bright objects found in the image with the known stars from the star catalog. Since a typical star catalog has more than 900 million stars, the star pattern search has to be made efficient. With estimates of the spacecraft's orientation available, the portion of the star catalog that needs to be accessed is reduced substantially. After developing the basic algorithm, the algorithm will be further optimized to utilize parallel computing graphical processor units (GPUs).

Task 6: Structural Integration

With the optics and electronics of SMOTESS in place, the structural components will be designed so that the spacecraft can survive launch forces and space conditions. Launch, which begins with engine ignition and ends with spacecraft separation, imposes a short-lived but highly stressful environment on the payload including structural loads, thermal transients, and rapid loss of pressure [2]. Launch vehicles need to be examined for their appropriateness in launching SMOTESS and the structural supports of SMOTESS designed accordingly.

Since the effect of gravity release is hard to measure on the ground, force actuators may be appropriately designed to fit the back of the mirror to make minute corrections to its figure once in orbit. This may be determined to be unnecessary since the mirrors of SMOTESS are relatively small. Mirror mounts and spacecraft trusses will be appropriately designed. Effects of temperature changes on optics alignment will be taken under account for structural material tradeoffs.

Task 7: Spacecraft Integration

Certain basic spacecraft parameters will be determined for integrating the SMOTESS payload. The coverage area of SMOTESS with one rotation every 6 to 24 hours requires the design of an attitude control system necessary for a dual spin spacecraft. The optimal orbit required for SMOTESS to function in its baseline design will determine the orbital transfers necessary and subsequently propulsion requirements.

c. Relationship and perceived impact of the research to the state of knowledge of the field

In recent years, there has been much progress in detection of NEAs, with over 90% of NEAs over absolute magnitude (H) 18 (~1 km diameter) detected. However, there are many NEAs smaller than 1 km diameter yet to be discovered. Table 3 shows the number of asteroids that remain to be discovered with their respective diameters.

Table 3: NEA detection summary with potentially hazardous asteroids (PHAs) in parenthesis [1].

***D* (m)**	**> 1000**	**1000 – 140**	**140 - 40**	**40 – 1**	**Total**
***H* (mag)**	< 17.75	17.75 - 22.0	22.0 - 24.75	> 24.75	
N Estimated	966 ± 45	~ 14,000	~ 285,000	?	
N Observed	900 (159)	4,709 (1239)	2,396	1,793	9,798 (1398)
O/E	93 % ± 4	~ 34 %	~ 1 %		

Over a span of ten years in a Venus-like orbit, running the raw data generated by Greenstreet et al. into our code shows that SMOTESS can detect ~93% of asteroids between the size of 140 m and 1000 m. More importantly, SMOTESS can discover ~56% of the asteroids between 40 m and 140 m, currently of which only 1% has been discovered. Through these discoveries, potential impact disasters can be averted and potential asteroids for resource extraction can be identified. The percentage of detection is likely to increase by giving the spacecraft the capability to reorient the rotational plane. A reorientation of the rotational plane will allow for greater sky coverage.

d. The general plan of work

The SMOTESS Management Plan begins from Matrix of Responsibilities which allocates the division of labor among the known Texas A&M University and ECAPS researchers. Shown in Table 4, this matrix shows how the PI is involved with each of the seven specific tasks. More specific details are highlighted in Appendix A. Texas A&M University is primarily responsible for optical development and structural configuration while ECAPS is primarily responsible for electronic and software development.

Table 4: Matrix of responsibilities.

Task	**Hyland (PI)**	**Kim**	**Tucker (CoI)**	**Satak (CoI)**	**Ge (CoI)**	**Smit**
1. Mission Req Defined	L			S	S	S
2. Optical Design	S	L	S		S2	
3. Electronic Design	S2	S	L		S2	
4. Optical and Electronic Integration	S2	S	L			S
5. Image Processing	S2	S		L		
6. Structural Integration	S2				L	S
7. Spacecraft Integration	L			S	S	S2

Key: L = Lead, S = Primary Support Collaboration, S2 = Team Support/Liaison

The work breakdown structure (WBS) in Figure 4 shows the team arrangements and assignments to support the overall task allocation. Dr. David Hyland, Professor of Aerospace Engineering at Texas A&M University serves as both PI and Project Manager. He enjoys support from the Texas A&M University Graduate Research Assistant Hyerim Kim and one other Graduate Research Assistant. Figure 5 shows the Milestones for both Year 1 and Year 2 necessary for project completion.

ECAPS provides the electronic engineer and image processing capabilities to complement the optical engineering capabilities provided by Texas A&M University. Roy Tucker serves as the main Electronics Engineer. Shen Ge serves as the System Engineer integrating all parts with the help of Engineer Klaus Smit. Dr. Neha Satak serves as the Digital Signal Processing Engineer responsible for on-board autonomous data processing to reduce data load before sending data to Earth.

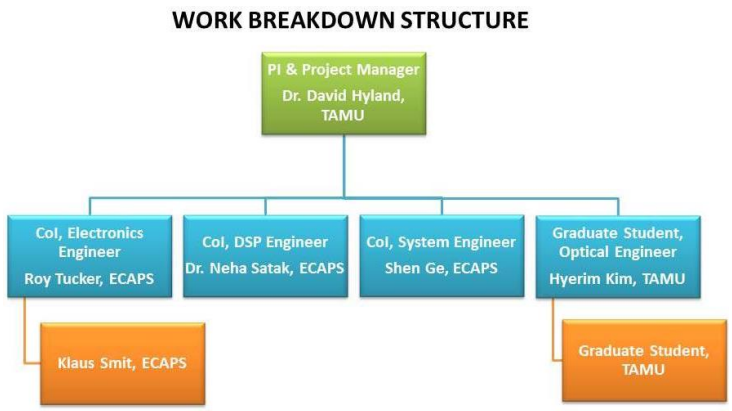


Figure 4: Texas A&M University and ECAPS work breakdown structure.

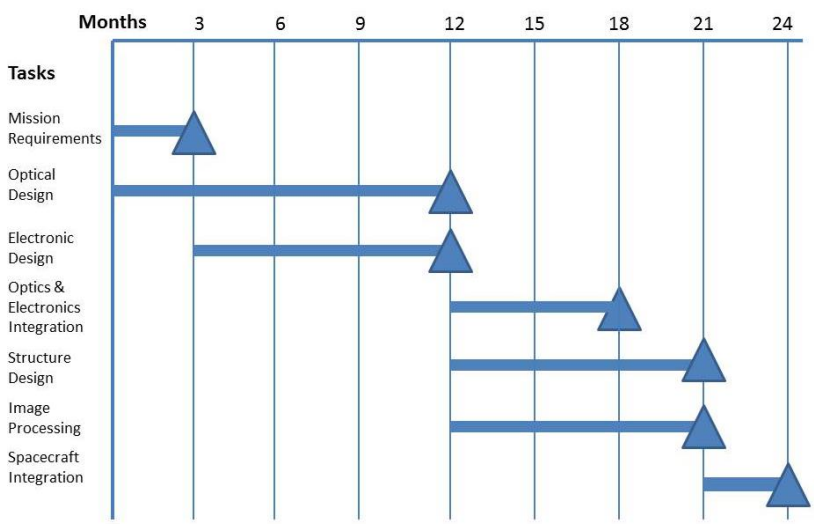


Figure 5: Milestone Chart for the SMOTESS project.

Month 0 to Month 6

The mission requirements for the SMOTESS payload to detect are identified and a model of the NEA population detectable by SMOTESS is simulated. Optical design of the telescope is initiated.

Month 6 to Month 12

Optical design of the telescope and electronic design of the scan-mode imaging with the CCD arrays are developed concurrently.

Month 12 to Month 18

Optical design and electronic design are integrated together. Image processing software is investigated and developed. Structural design of the payload based on the mass, size, and power requirements of the telescope and electronic array is developed.

Month 18 to Month 24

Image processing software and structural design will be concluded. Upon conclusion, the entire payload's mass, power, and size requirements will be confirmed. These will include the telescope, electronics, and structure as well as any other components.

e. Data Sharing Plan

Submission of peer-reviewed conference and journal publications is planned for three times per year for both Year 1 and the Year 2 options. Since the final aim of SMOTESS is to make new NEA discoveries, provision must be made for reporting astrometric data to the Minor Planet Center (MPC). This would require a special format as is done for other spacecraft surveys and radar observations (provides really good range and Doppler velocities). We plan to negotiate with the MPC to develop the preferred report format that could be emailed to them. The Co-I, Roy Tucker has discovered hundreds of asteroids, including co-discovery of Apophis, and is deeply experienced in the discovery reporting process.

4. References and Citations

[1] "Near Earth Asteroids." *International Astronomical Union*. http://www.iau.org/public/nea

[2] *The Design and Construction of Large Optical Telescopes.* Ed. Bely, Pierre Y. New York: Springer, 2003. *Springer-Verlag*.

[3] Bottke Jr., et al. "Debiased Orbital and Absolute Magnitude Distribution of the Near-Earth Objects." *Icarus* Vol. 156, Iss. 2: Apr 2002, 399-433. ScienceDirect Database.

[4] DeBenedictis, E., Huang, T., and Hong, C.”A Novel Approach to Asteroid Identification Using Image Processing of Existing Data.” Report. <http://www.debenedictis.org/erika/ISEFreport1v1.pdf>

[5]Greenstreet, S., Ngo, H., and Gladman, B. “The Orbital Distribution of Near-earth Objects inside Earth’s Orbit.” *Icarus* Vol. 217, Iss. 1: Jan 2012, 355-366. ScienceDirect Database.

[6] Leinert, C., et al. “1997 Reference of Diffuse Night Sky Brightness.” *Astronomy and Astrophysics* Feb 8, 1997. <http://trs-new.jpl.nasa.gov/dspace/bitstream/2014/22995/1/97-1547.pdf>

[7] Tucker, Roy. “MOTESS: The Moving Object and Transient Event Search System.” *The Astronomical Journal* Vol. 134: Oct 2007, 1483-1487.

5. Biographical Sketches

BIOGRAPHICAL SKETCH
DR. DAVID HYLAND
PRINCIPAL INVESTIGATOR AND PROJECT MANAGER

Dr. David C. Hyland

Education

Ph.D., Aeronautics and Astronautics, Massachusetts Institute of Technology, 1973
M.S., Aeronautics and Astronautics, Massachusetts Institute of Technology, 1971
B.S., Aeronautics and Astronautics, Massachusetts Institute of Technology, 1969

Professional Experience

February 1, 2005 - Present : Director of Space Science and Space Engineering Research, Texas A&M University Professor of Aerospace Engineering, College of Engineering, holder of the Wisenbaker Chair of Engineering and Adjunct Professor of Physics, College of Science.
September 2003 – March 2005: Associate Vice Chancellor of Engineering, Texas A&M, Deputy Director of TEES (Texas Engineering Experiment Station), Associate Dean of the Dwight Look College of Engineering, Professor of Aerospace Engineering, College of Engineering, holder of the Wisenbaker Chair of Engineering and Adjunct Professor of Physics, College of Science.
May, 1996 – August, 2003: Professor and Chairman of the Aerospace Engineering Department of the University of Michigan, Ann Arbor.
February 1992 – April 1996: Senior Scientist, Aerospace Systems Division, Harris Corporation and member of the Senior Staff of the Vice President of Engineering of the Aerospace Systems Division.
August 1983 – January 1994: Director of the Structural Control Group, Government Aerospace Systems Division of Harris Corporation.
November 1974 - July 1983: Technical Staff Member, the MIT Lincoln Laboratory.
November 1973 – October 1973: Technical staff member, Cambridge Collaborative, Inc.
September 1969 - November 1973: Research Associate at the MIT Lincoln Laboratory.

Grants and Sponsored Research

TAMU Support for the AFRL/RV ICI Spaceflight Experiment. March 2012 - Feb. 2013. AFRL/RV. $90K.

Proof-of-Concept Experiment for Intensity Correlation Imaging of GEO Objects. September 2010 to September 2012. $145K

Intensity Correlation Imaging of Geostationary Inputs: A Multi-Spectral Receiver, DARPA, November 2010 to August 2012. $225K

"Apophis Mitigation Technology Demonstration for Space Flight on KACST Satellite". PI: D. C. Hyland, $1450K, Oct. 2012 – July 2013, King Abdulaziz City for Science and Technology (KACST), Kingdom of Saudi Arabia.

Intensity Correlation Imaging for Space Situational Awareness, AFRL/VS, June 2005 – present, $253K.

Self-Reliant, High Resolution Imaging Using Formation Flying Spacecraft, NASA, CETDP Mission, 2/15/2001 – 2/14/2006, $1,928 K (D. Hyland, PI, P. Kabamba and D. Scheeres, Co-Is)

Spacecraft Formation Flying to Optimize Imaging Quality in Long

Baseline Interferometry, NASA/JPL , PI, Sept. 1999 – Sept. 2001, $120 K.

Recent Publications (From 2005)

D.C. Hyland, "Exo-Planet Detection Via Stellar Intensity Correlation Interferometry, Techniques and Instrumentation for Detection of Exoplanets II, SPIE International Symposium on Optics & Photonics, SPIE paper no. 5905-37, 31 July – August 4, 2005, San Diego, California.

I. I. Hussein, D. J. Scheeres, and D. C. Hyland. "Optimal formation design for imaging and fuel usage". *Journal of Guidance, Control and Dynamics,* Vol.6, May, 2007.

D. C. Hyland, "Efficient, Passively Orbiting Constellations for High Resolution Imaging of Geosynchronous Objects". 2007 AAS/AIAA Astrodynamics Specialist Conference, Mackinac Island, Michigan, August 19-23, 2007.

Tonetti, S., Hyland, D., and Mortari, D. "The Flower Formation Flying. Part II; Application," Paper AAS 08-186 of the 2008 AAS Space Flight Mechanics Meeting Conference, Galveston, TX, January 27-31, 2008.

D.C. Hyland, H.A. Altwaijry, S. Ge, R. Margulieux, J. Doyle, J. Sandberg, B. Young, X. Bai, J. Lopez, N. Satak, "A Mission Template for Exploration and Mitigation of Potentially Hazardous Near Earth Asteroids" International Symposium on Hazardous Near-Earth Asteroids, Malta. October, 2009.

D. C. Hyland, D. Khussainov, and H. Kim. "Design of a Space-borne, Mid-IR Exoplanet Imager" American Astronomical Society 217th Meeting. Seattle, 9-13, January 2011.

D. C. Hyland, and H. A. Altwaijry. "Detection and Characterization of Near Earth Asteroids Using Stellar Occultation". 2013 AAS/AIAA Astrodynamics Specialist Conference, Hilton Head, S.C., August 2013.

R. Trahan III, and D. C. Hyland, "Mitigating the effect of noise in the hybrid input-output method of phase retrieval", *IEEE Trans. Appl. Optics*, vol. 52, Issue 13, 2013 pp. 3031-3037.

BIOGRAPHICAL SKETCH
HYERIM KIM
CO-INVESTIGATOR AND OPTICAL ENGINEER

Hyerim Kim is a Ph. D student in Aerospace engineering at Texas A&M University and received her B.S. in Astronomy and Space science and B.E. in Mechanical informatics engineering from Kyung Hee University, South Korea in 2009. Her research interests include optical imaging and control, space dynamics and design.

EDUCATION

Ph.D.	Present	Aerospace Engineering	Texas A&M University
BS	2009	Astronomy and Space Science	Kyung Hee University
BE	2009	Mechanical Information Engineering	Kyung Hee University

AWARDS

1. **2013 Amelia Earhart Fellowship** presented by Zonta International.
2. **2004-2005, 2008 Scholarship in Recognition of Outstanding Academic Excellence**

ACHIEVEMENT

Texas A&M University (2009-Present) **Research Assistant**
- Designing optimal trajectories to Near-earth Asteroids (NEAs) by using invariant manifolds of the Patched Circular Three-Body Problem (PC3BP) with small delta-v.
- Developing payload and integrating with spacecraft mission for the Apophis Mitigation Technology project.

Teaching Assistant (2011 – Present)
- Taught senior level design courses on long-duration interplanetary spacecraft mission with Excalibur Almaz and several capstone senior design courses on multiple satellites requiring rendezvous and docking.

Project Manager (2010 – 2011)
- Organized system architecture and coded imaging simulations for an exo-planet finder proposal.

Conference Organizer (2011)
- Co-organized the International Symposium of Asteroid Mitigation and Exploration.

Kyung Hee University (2007-2009) **Research Assistant**
- Worked in the Space Science and Technology Laboratory on Multiple Flux System in Halo CME Source Region.

RELEVANT SKILLS

Programming: MATLAB, C++, LABVIEW, IDL
Design: AutoCAD, CATIA, Pro/Engineer
Office: MS Word, MS Excel, MS Powerpoint, LaTeX

BIOGRAPHICAL SKETCH
ROY TUCKER
CO-INVESTIGATOR AND ELECTRONIC ENGINEER

EDUCATION

MS	Scientific Instrumentation	University of California, Santa Barbara
BS	Physics	Memphis State University

RELEVANT EXPERIENCE IN LAST 10 YEARS

Imaging Technology Laboratory, Tucson, Arizona 9/02 to present

Senior Engineer - Characterized the performance of scientific CCD imaging arrays.

National Solar Observatory, Tucson, Arizona 10/97 to 5/02

Senior Engineer - Participated in developing the high-resolution upgrade of the GONG (Global Oscillation Network Group) helioseismic instrumentation.

Phase Shift Technology, Tucson, Arizona 2/95 to 10/97

Electronics Engineer - Chief electronics engineer for a company specializing in optical metrology. This work is heavily involved with video digitizers, real-time video processors, and electrical noise reduction in E-O instrumentation systems.

Southwest Cryostatics, Albuquerque, New Mexico and Tucson, Arizona 12/93 to 12/00

Sole Proprietor, Entrepreneur - Marketed documentation and parts for a low-cost instrument-quality astronomical CCD camera.

Applied Technology Associates, Albuquerque, New Mexico 10/92 to 12/94

Electro-optical Instrumentation Engineer - Provided expertise in the application of electro-optical technology to problems in high resolution astronomical imaging, computer interfacing, data acquisition, and process control. Acquired familiarity with wavefront sensing and adaptive optics.

RELEVANT ACCOMPLISHMENTS

- Winner of the first three Benson Awards for the amateur discovery of Near-Earth Asteroids

- Winner of two Edgar Wilson Awards for the amateur discovery of comets

- Recipient of a Eugene Shoemaker Award (2002) in support of his asteroid search efforts

- Co-discoverer of Earth-threatening asteroid (99942) Apophis, also known as 2004 MN4

BIOGRAPHICAL SKETCH
SHEN GE
SYSTEMS ENGINEER

Shen Ge received his Masters in aerospace engineering from Texas A&M University in 2011 August and received his undergraduate with a dual major in aerospace engineering and physics with Magna Cum Laude in 2008 December from Georgia Institute of Technology. His background is in space design, space physics, and system engineering.

EDUCATION

MS	2011	Aerospace Engineering	Texas A&M University
BS	2008	Aerospace Engineering	Georgia Institute of Technology
BS	2008	Physics	Georgia Institute of Technology

ACHIEVEMENTS

Experimental Center for Applied Physical Systems, LLC (2012-Present) **Co-owner**
- Researching on asteroid mining technology and space telescope technology.
- Managing the Apophis Mitigation Technology (AMT) project.

Scientific Preparatory Academy for Cosmic Explorers, Inc. (2012-Present) **Co-founder**
- Launched one international space conference on the Isle of Man and one international space workshop on the Canary Islands.
- Working on developing space education modules, i.e., short-term space design courses taught in foreign universities.

Excalibur Almaz (2012) **Research Intern**
- Designed cislunar spacecraft power and life support systems.
- Simulated drag and heating of tethers for reentry vehicles.

Texas A&M University (2009-2012) **Research Assistant**
Developed payload and simulated asteroid deflection payload's performance.
Co-wrote Apophis Near Earth Asteroid Earth Impact Mitigation Proposal.
Proposal Writer
2012 New Space Business Plan Competition: Citizen Spacecraft Design Platform.
2011 European Satellite Navigation Competition/
Conference Organizer (2011)
2011 Co-organized the International Symposium of Asteroid Mitigation and Exploration.

RELEVANT SKILLS

Programming: MATLAB, C++, PHP/mySQL, HTML, JavaScript
Design: Solid Edge, SolidWorks
Office: MS Word, MS Excel, MS Powerpoint, MS Publisher

BIOGRAPHICAL SKETCH
NEHA SATAK
CO-INVESTIGATOR AND DIGITIAL SIGNAL PROCESSING ENGINEER

Neha Satak is a recent doctoral graduate from the Department of Aerospace Engineering at Texas A&M University. She co-founded in January 2012 the Scientific Preparatory Academy for Cosmic Explorers (SPACE), a nonprofit company located on the Isle of Man aiming to be an undergraduate space university and also co-founded in July 2012 the Experimental Center for Applied Physical Systems (ECAPS), LLC, a research and development company in College Station, TX.

EDUCATION

Ph.D.	2013	Aerospace Engineering	Texas A&M University
M.S.	2008	Aerospace Engineering	Indian Institute of Science
B.E.	2006	Electronics and Communication Engineering	Rajasthan University

ACHIVEMENTS

- **Chief Operating Officer at the Scientific Preparatory Academy for Cosmic Explorers (SPACE), Isle of Man, UK.**
    - **Organized two conference**
    - **Work on Space education initiatives**
- **Partner, ECAPS, LLC.**
    - **Research on asteroid mining economic feasibility**
    - **Research on developing technology to deflect asteroids**
- **Internship, Excalibur Almaz, Houston, Texas**
    - **Modeling reentry vehicle**
    - **Reentry trajectory optimization**

AWARDS

1. **2011 Amelia Earhart Fellowship** presented by Zonta International.
2. **2009 NASAS Gold Medal for Best Master's Thesis** presented by the Indian Institute of Science, Bangalore.

RELEVANT PROJECTS

Image Processing based Robot Navigation
2005-2006, Department of Electronics & Communication Engg., Rajasthan University.

Deflection of the Asteroid Apophis
2009-2011, Department of Aerospace Engg., Texas A&M University.

Navigation about an Asteroid
2010-2011, Department of Aerospace Engg., Texas A&M University.

BIOGRAPHICAL SKETCH
KLAUS SMIT

Klaus Smit is a Ph. D student in Physics at Texas A&M University and received his B.S. in Physics from the University of Florida in 2011. His research background includes optical systems, image processing, simulations, and superconducting magnets.

EDUCATION

Ph.D.	Present	Physics	Texas A&M University
BS	2009	Physics	University of Florida

AWARDS

Bridge to Doctorate Fellowship, August 2011– August 2013

- Award consists on a $30,000 annual stipend and a $9,000 cost of education allowance for attending Texas A&M University, funded through the NSF.

Florida-Georgia Louis Stokes for Minority Participation, August 2010 – May 2011

- Award consists of $1,500 and professional development activities to help prepare participants for graduate school.

REVELANT EXPERIENCE

Dr. **Peter McIntyre Research Group at Texas A&M University,** College Station, Texas
Research Graduate Assistant, May 2011 – Present

- Designed a cold finger cryostat for converting a warm bore NMR into a cold bore for measuring super-current transport in samples of Bi-2212 textured powder wire.
- Contributed to creating a manufacturing process for highly textured Bi-2212 superconductor ribbon tape, by encapsulating it with silver foils and making a wire by assembling it into a "Jelly-roll" configuration.

LIGO (Laser Interferometer Gravitational Waves Observatory), University of Florida, Gainesville, FL

Lab Assistant, September 2009 – April 2011

- Developed an adaptive control system to compensate for distortions on a laser beam due to thermal lensing caused by the absorption of the laser power.
- Extracted beam profiles by acquiring data with gigabit Ethernet camera and processing the data by interfacing Matlab and Labview

RELEVANT SKILLS

Programming: MATLAB, C++, LABVIEW
Design: AutoCAD, Inventor
Office: MS Word, MS Excel, MS PowerPoint

6. Current and Pending Support

CURRENT SUPPORT

Dr. David C. Hyland

Title of award: Near Earth Object Studies

Names of PIs: D. C. Hyland, Shen Ge

Sponsoring agency: King Abdulaziz City for Science and Technology (KACST), Space Research Institute (SRI)

Point of Contact: Dr. Haithem A. Altwaijry, Director, SRI

haltwaijry@kacst.edu.sa, +966 1 4814532

Performance period: September 2012 through September 2013.

Total Budget: $996,956.00

Commitment by PI: 2 person-months per year

Hyerim Kim

Title of award: Amelia Earhart Fellowship 2013-2014

Sponsoring agency: Zonta International Foundation

Point of Contact: Martina Gamboa, Programs Coordinator, Zonta International

mgamboa@zonta.org, +1 630 928 1400

Performance period: September 2013 through April 2014

Total Budget: $10,000.00

Commitment: 8 person-months per year

Title of award: Near Earth Object Studies

Sponsoring agency: King Abdulaziz City for Science and Technology (KACST), Space Research Institute (SRI)

Point of Contact: Dr. Haithem A. Altwaijry, Director, SRI

haltwaijry@kacst.edu.sa, +966 1 4814532

Performance period: September 2012 through September 2013.

Total Budget: $996,956.00

Commitment: 3 person-months per year

Shen Ge

Title of award: Near Earth Object Studies

Names of PIs: D. C. Hyland, Shen Ge

Sponsoring agency: King Abdulaziz City for Science and Technology (KACST), Space Research Institute (SRI)

Point of Contact: Dr. Haithem A. Altwaijry, Director, SRI

haltwaijry@kacst.edu.sa, +966 1 4814532

Performance period: September 2012 through September 2013.

Total Budget: $996,956.00

Commitment by Co-PI: 3 person-months per year

Roy Tucker

Title of award: Undisclosed

Sponsoring agency: University of Arizona Grants and Contracts

Point of Contact: Senior Vice President for Research, University of Arizona

+1 520 621 3513

Performance period: Indefinite

Total Budget: Undisclosed

Commitment: 12 person-months per year

Neha Satak

Title of award: AFOSR Optimization and Discrete Mathematics Program

Sponsoring agency: AFRL

Point of Contact: Kevin L. Brink, Research Engineer, AFRL/RWWI

kevin.brink2@eglin.af.mil, +1 850 882 4600

Performance period: June 2013 through June 2014

Total Budget: Undisclosed

Commitment: 12 person-months per year

PENDING SUPPORT

Personnel	Source	Short Title	Position	Performance Period	Total Budget	Start Date	Finish Date
Hyland	NASA STRO-ESI	Asteroid Occultation	PI	3 person-months/year	$175,000	1-Sep-2013	30-Aug-2015
Kim	NASA STRO-ESI	Asteroid Occultation	CoI	3 person-months/year	$175,000	1-Sep-2013	30-Aug-2015
Ge	NASA STRO-ESI	Asteroid Occultation	CoI	2 person-months/year	$75,000	1-Sep-2013	30-Aug-2015
Tucker	NASA STRO-ESI	Asteroid Occultation	CoI	2 person-months/year	$75,000	1-Sep-2013	30-Aug-2015
Satak	NASA STRO-ESI	Asteroid Occultation	CoI	2 person-months/year	$75,000	1-Sep-2013	30-Aug-2015

7. Statements of Commitment

We have letters from everyone confirmed electronically through NSPIRES.

8. Budget Justification

8a. TEES Budget

The type of award desired, is a contract to Texas Engineering Experiment Station (TEES) with a subcontract to ECAPS LLC. We consider TEES and ECAPS in turn and detailed information are shown in Appendix A.

TEES

TEES personnel comprise the PI, Prof. Hyland and graduate student, Hyerim Kim for both years. For detailed information, please consult Appendix A.

For year 1, the PI, who has a deep understanding of optics, will supervise Tasks 1 – 3 with a focus on Task 1 and Task 2. The Co-I Mr. Roy Tucker of ECAPS LLC, who has substantial optical electronics experience, will be working closely with Mrs. Kim of TEES on Task 3. Mrs. Kim, a graduate student under the PI, has an astronomy and optics background and will work with both the PI and Mr. Roy Tucker on designing the optics and electronics. A less experienced graduate student for low-level programming tasks will aid Mrs. Kim on Task 2 and Task 3.

For activities during year 2, the PI maintains the same supervisory level of effort as in year 1 over Tasks 4 – 7 with a focus on Task 5 and Task 7. Mrs. Kim will continue her work in optics and electronics by integrating the two together for Task 4. Furthermore, Mrs. Kim is experienced in image processing for astronomical purposes and her role is essential for developing an autonomous onboard image processing algorithm for Task 5. A less experienced graduate student for low-level programming tasks will aid Mrs. Kim on Task 4 and Task 5.

Aside from general purpose equipment such as computers, the PI foresees no facilities and equipment expenditures called for in this proposal.

Regarding the subcontract expense, the rationale is explained in the following sub section.

ECAPS LLC

ECAPS personnel comprise Mr. Roy Tucker, Mr. Shen Ge, and Dr. Neha Satak. The reason for choosing ECAPS personnel for these tasks is that their experience and expertise are uniquely suited to the present proposal. Mr. Roy Tucker is a very experienced asteroid hunter (and co-discoverer of Asteroid Apophis) as well as an experienced astronomer and instrumentation engineer. His assistance with the electronics work (Task 3) and the integration of optics and electronics design (Task 4) will be of the greatest benefit.

Mr. Ge's graduate work included statistical characterization of NEAs and the invention of a new mitigation method. Dr. Satak collaborated with Mr. Ge in modeling asteroid distributions. Furthermore, in their experience as graduate students at Texas A&M, both individuals have experience in space mission design – in particular, a joint effort with NASA Ames to design the Apophis Exploration and Mitigation Mission. These capabilities are directly relevant to Task 1, 6, and 7. Furthermore, Dr. Satak has substantial experience in image processing starting from

undergraduate course work and later in a project on processing the images of asteroids which leads her to be a capable counterpart to work with Mrs. Hyerim Kim for Task 5.

Referring to Appendix A which details the levels of effort for year 1, we estimate that quarter-time effort by Mr. Ge and Dr. Satak should be sufficient to complete Task 1. For Task 1, they will be assisted in lower-level programming tasks by engineer Klaus Smit working half-time. Mr. Ge as a systems engineer will also be doing supporting work on Tasks 2 and 3 to understand those developments in parallel. Mr. Tucker will focus on designing the electronic controllers (Task 3) with supporting work for the optics (Task 2).

Efforts in the second year are shown in Appendix A. Mr. Tucker will be heavily involved in the integration of optics design with electronics design (Task 4). He will be working with Mrs. Kim on Task 4 and the ECAPS engineer Klaus Smit. We estimate quarter time by Mr. Ge on Task 6 with supporting role on Task 7, and quarter time by Dr. Satak on Task 5 with supporting role on Task 7.

8b. Budget Details

The budget detail for TEES is shown in Table 6. The PI is listed for 3 person-months per year, both years at an annual salary of approximately $58,353.00. The amount of time devoted to the project by all labor categories is listed. Fringe benefits for faculty, staff and students are indicated, as are group insurance rates. Conference registration and travel are estimated as described above and there is no equipment cost. Other direct costs, not including MTDC, comprise the subcontract to ECAPS. ECAPS budget detail is given separately below. F&A rates negotiated with DHHS are 45.5% as indicated. The name, address, and telephone number of the Federal agency official having cognizance are: Department of Human Services, Washington, D.C., Theodore Foster. 214-767-3411. As indicated, the total estimated funding requested of the Government is $500,000.00.

BUDGET DETAIL - SMOTESS
1314089

Agency: NASA
Proposed Term: 1/Sep/13-30/Aug/15

							Year 1	Year 2	Summary
A. Senior Personnel*									
PI: David Hyland	Yr 1				3.00 P-Mos per year		$58,353	$60,104	$118,457
	Yr 2				3.00 P-Mos per year				
						Total Senior Personnel	$58,353	$60,104	$118,457
B. Other Personnel*									
Graduate Student - PhD	Yr 1	1	@		6.00 P-Mos per year		$22,860	$23,545	$46,405
	Yr 2	1	@		6.00 P-Mos per year				
Graduate Student - PhD	Yr 1	1	@		2.75 P-Mos per year		$9,902	$13,616	$23,518
	Yr 2	1	@		3.67 P-Mos per year				
						Total Other Personnel	$32,762	$37,161	$69,923
						Total Salaries and Wages	**$91,115**	**$97,265**	**$188,380**
C. Fringe Benefits									
Faculty and Staff at						17.4%	$10,153	$10,458	$20,611
Students at						2.30%	$754	$855	$1,609
						Total Fringe Benefits	**$10,907**	**$11,313**	**$22,220**
D. Group Insurance									
PI: David Hyland	Yr 1	3.00	P-Mos	@	$474	per month	$1,422	$1,422	$2,844
	Yr 2	3.00	P-Mos	@	$474	per month			
Graduate Student - PhD	Yr 1	12.00	P-Mos	@	$188	per month	$2,256	$2,256	$4,512
	Yr 2	12.00	P-Mos	@	$188	per month			
Graduate Student - PhD	Yr 1	5.50	P-Mos	@	$188	per month	$1,034	$1,380	$2,414
	Yr 2	7.34	P-Mos	@	$188	per month			
						Total Group Insurance	**$4,712**	**$5,058**	**$9,770**
						Total Salaries, Wages, Fringe Benefits & Group Insurance	**$106,734**	**$113,636**	**$220,370**
E. Permanent Equipment									
						Total Permanent Equipment	**$0**	**$0**	**$0**
F. Travel									
						Total Travel	**$0**	**$0**	**$0**
G. Other Direct Costs - Included in MTDC									
First $25,000 of Sub-Recipient						Experimental Center for Applied Physical Sys	$25,000	$0	$25,000
						Total Other Direct Costs - *Included in MTDC*	**$25,000**	**$0**	**$25,000**
H. Other Direct Costs - Not included in MTDC									
Sub-Recipient after first $25,000						Experimental Center for Applied Physical Sys	$50,000	$75,000	$125,000
Graduate Student Tuition and Allowable Fees **							$8,327	$9,660	$17,987
						Total Other Direct Costs - *Not* *Included in MTDC*	**$58,327**	**$84,660**	**$142,987**
I. Modified Total Direct Costs (MTDC)							**$131,734**	**$113,636**	**$245,370**
J. Total Direct Costs							**$190,061**	**$198,296**	**$388,357**
K. Facilities & Administrative Costs (F&A)									
Rate negotiated with the DHHS					45.5%	MTDC	**$59,939**	**$51,704**	**$111,643**
L. Total Project Costs							**$250,000**	**$250,000**	**$500,000**

* Includes a 3% salary escalation beginning September 1 of each year
**Includes a yearly 5% escalation on tuition and allowable fees

Table 6: TEES Budget detail

If selected, we affirm that we will comply with the policy of the Office of Management and Budget according to the following terms:

a) The students are engaged in activities necessary to the sponsored agreement.
b) Tuition remission and other support are provided in accordance with established educational institutional policy and consistently provided in a like manner to students in return for similar activities conducted in nonsponsored as well as sponsored activities.
c) During the academic period, the student is enrolled in an advanced degree program at the institution or affiliated institution and the activities of the student in relation to the NASA-sponsored research project are related to the degree program.
d) The tuition or other payments are reasonable compensation for the work performed and are conditioned explicitly upon the performance of necessary work.
e) The student aid costs conform to the institution's practice to similarly compensate students in nonsponsored as well as sponsored activities.

8c. ECAPS Budget

EXPERIMENTAL CENTER FOR APPLIED PHYSICAL SYSTEMS LLC

SUBRECIPIENT ATTACHMENT FOR PROPOSAL WITH

Texas A&M University

ENTITLED

A Novel Telescope for NEO Detection: Space Moving Object and Transient Event Search System (SMOTESS)

ECAPS CONTACT

Name: Shen Ge

Office/Fax: (979) 703-6376

Mobile: (770) 617-4046

STATEMENT OF WORK

Experimental Center for Applied Physical Systems LLC (henceforth referred to as ECAPS) has extensive experience in space physics modeling as well as telescope electronics and software. The Space Moving Object and Transient System (SMOTESS) will be a space adaptation of the existing ground-based Moving Object and Transient System (MOTESS) already designed and used by ECAPS Electronics Engineer and Co-Investigator Roy Tucker.

ECAPS will be taking the lead in two areas:

1. Develop the NEO population model to determine the efficacy of detecting NEAs for various orbits.
2. Design the image processing algorithm and software capable of autonomous detection of near earth asteroids integrated with the SMOTESS instrument.

Furthermore, ECAPS will work with Texas A&M University on estimating the performance parameters of the space telescope (such as limiting magnitude and scan rate) to give an accurate portrayal of SMOTESS's capabilities.

The work conducted by ECAPS will be located at ECAPS's office at 707 Texas Avenue Suite 222, College Station, TX 77840-1967.

The Work Breakdown Structure and Milestones Chart are depicted in the Figures 6 and 7.

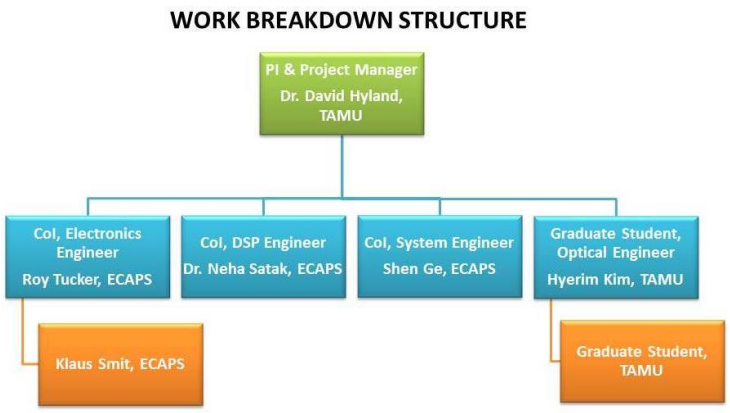


Figure 6: Work Breakdown Structure (WBS) of the SMOTESS project.

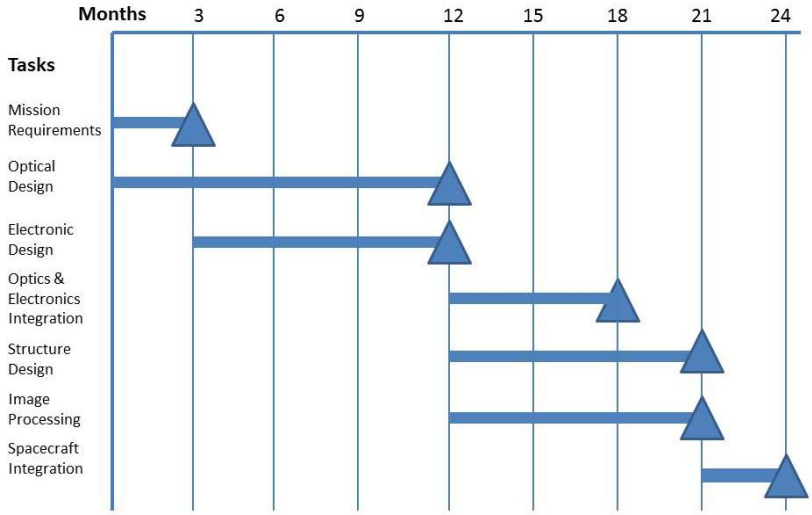


Figure 7: Milestones Chart for the SMOTESS project.

BUDGET JUSTIFICATION

The stated budget of $150,000 for ECAPS is based on the estimated rates of four professional engineers, one with exceptional experience, with respective time fractions over the two-year period of the project. Allocations for the first and second year are shown in Table 7 and Table 8 respectively.

Table 7: Allocation for the first year.

Task	**Personnel**	**Position**	**Entity**	**Time Fraction**	**Total Hours**	**Labor Rate (USD/hour)**	**Labor Total (USD)**
System Engineering	Shen Ge	CoI	ECAPS	0.250142	520.2952	$27.10	$14,100.00
Electronic Engineering	Roy Tucker	CoI	ECAPS	0.25	520	$60.00	$31,200.00
Digital Signal Processing	Dr. Neha Satak	CoI	ECAPS	0.250142	520.2952	$27.10	$14,100.00
Engineering	Klaus Smit	Engineer	ECAPS	0.5	1040	$15.00	$15,600.00
Net Labor Cost to NASA							**$75,000.00**

Table 8: Allocation for the second year.

Task	**Personnel**	**Position**	**Entity**	**Time Fraction**	**Total Hours**	**Labor Rate (USD/hour)**	**Labor Total (USD)**
System Engineering	Shen Ge	CoI	ECAPS	0.250142	520.2952	$27.10	$14,100.00
Electronic Engineering	Roy Tucker	CoI	ECAPS	0.25	520	$60.00	$31,200.00
Digital Signal Processing	Dr. Neha Satak	CoI	ECAPS	0.250142	520.2952	$27.10	$14,100.00
Engineering	Klaus Smit	Engineer	ECAPS	0.5	1040	$15.00	$15,600.00
Net Labor Cost to NASA							**$75,000.00**

Currently, Texas A&M possesses one computer lab and one research lab to develop SMOTESS. ECAPS possesses an office with computers and software to design models, analyze data, and develop software for SMOTESS development. Furthermore, collaboration between ECAPS

personnel and Texas A&M personnel is readily provided by the close proximity of the facilities and video conferencing capability provided by ECAPS. No funding from STRO-ESI will go into purchasing software or hardware.

Appendix A: Detailed Work Plan

List of Tasks	**Team Member(s)**	**No. Weeks Per Year**	
		Year 1	**Year 2**
Task 1: Define mission requirements.			
Sub-Task 1a: Develop simplified model of SMOTESS instrument for detection	Hyland/Smit/Ge	**1/2/1**	
Sub-Task 1b: Spacecraft trajectory simulation	Satak/Smit	**9/2**	
Sub-Task 1c: Simulation of asteroid detection using SMOTESS model and spacecraft trajectory	Ge/Satak/Smit	**2/3/8**	
Sub-Task 1d: Integrating code from sub-task 1b and 1c	Hyland/Ge/Smit	**2/1/4**	
Sub-Task 1e: Generate detection profile for different orbits of the spacecraft	Hyland/Ge/Smit	**2/2/7**	
Task 2: Design optical components.			
Sub-Task 2a: CCD physical, electrical, and optical characteristics	Kim/Grad/Tucker	**8/8/6**	
Sub-Task 2b: CCD enclosure in telescope	Hyland/Grad/Ge	**2/8/1**	
Sub-Task 2c: Telescope control including primary mirror and latitude and axial adjustments	Hyland/Kim/Grad	**6/8/8**	
Sub-Task 2d: Primary mirror material and structure	Ge	**3**	
Task 3: Design CCD camera controllers.			
Sub-Task 3a: Clock generator	Kim/Grad/Tucker	**4/8/1**	
Sub-Task 3b: Video processor	Kim/Grad/Tucker	**3/8/2**	
Sub-Task 3c: Control logic	Kim/Grad/Tucker	**3/8/2**	
Sub-Task 3d: Power supplies	Grad/Tucker/Ge	**4/3/3**	
Year 1 Total Person-Weeks		**153**	
Task 4: Intergrate optics with electronics.			
Sub-Task 4a: Investigate spinning and torquing spacecraft configurations	Kim/Tucker/Smit		**2/5/6**
Sub-Task 4b: Determine optimal CCD clocking rate as a function of spin rate of the spacecraft	Kim/Grad		**8/8**
Sub-Task 4c: Develop telescope focus and collimation mechanism	Grad/Tucker		**4/8**
Task 5: Write image processing algorithm.			
Sub-Task 5a: Develop algorithm to find the orientation and pointing direction of a CCD image	Hyland/Kim/Grad/Satak		**3/4/8/4**
Sub-Task 5b: Extract objects from the pictures and process them for asteroid detection	Hyland/Kim/Grad		**4/6/8**
Sub-Task 5c: Create mechanism to identify false detections and orbit determination	Kim/Grad/Satak		**6/8/3**
Sub-Task 5d: Identify GPUs and other hardware	Grad/Satak		**8/2**

Sub-Task 5e: Code the algorithm to use parallel computing	Grad/Satak		**8/2**
Task 6: Design structure to support optics and electronics.			
Sub-Task 6a: Force actuator design	Ge/Smit		**2/5**
Sub-Task 6b: Mirror mounts and telescope truss design	Ge/Smit		**2/5**
Sub-Task 6c: Thermal design	Ge/Smit		**4/4**
Sub-Task 6d: Support for camera and radiation shielding	Ge/Smit		**4/4**
Task 7: Integrate SMOTESS with a spacecraft.			
Sub-Task 7a: Structural requirements	Hyland/Smit		**3/1**
Sub-Task 7b: Spin axis monitoring and control	Hyland/Satak		**2/1**
Sub-Task 7c: Spin rate monitoring (Pointing accuracy)	Hyland/Satak		**2/1**
Sub-Task 7d: Thermal requirements	Ge/Smit		**1/1**
Year 2 Total Person-Weeks			**157**
Total Person-Weeks			**310**